

Asadullah Bin Rahman



galib.hstu.cse17@gmail.com



asadullahgalib007.github.io



in/asadullah007



Google Scholar

Research Interests: Image Processing, Machine Learning, Quantum Computing

Employment

- 07/25 – Present ♦ **Lecturer, Computer Science and Engineering**
World University of Bangladesh, Dhaka, Bangladesh
- 07/23 – 07/25 ♦ **Research Assistant**
IoThink Lab, Department of Computer Science and Engineering, HSTU
- 03/23 – 09/23 ♦ **Lecturer, Computer Science and Engineering**
Govt. Shahid Akbar Ali Science and Technology College (Affiliated to HSTU), Bangladesh

Education

- 07/23 – 07/25 ♦ **MSc in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
CGPA : 3.547/4.00
- 01/17 – 12/22 ♦ **BSc (Engineering) in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
CGPA : 3.31/4.00

Skills

- Quantum Computing ♦ Proficient in Qiskit, Cirq, PennyLane, Classiq; Expertise in Quantum Algorithms, Quantum Machine Learning, Quantum Key Distribution
- Machine Learning ♦ Proficient in NumPy, Pandas, SciPy, Matplotlib, OpenCV, PyTorch, TensorFlow; Expertise in Image Processing, Computer Vision
- Programming ♦ C++, Java, Python; Data Structures, Algorithms, Databases; Solved 300+ problems (CodeForces, HackerRank, UVA)
- Languages ♦ Bangla (Native), English (IELTS Academic: **6.5**, CEFR Level: B2)
- Others ♦ Linux, Git, LaTeX

Research Publications

Conference Proceedings

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, “Mitigating Noise from Biomedical Images Using Wavelet Transform Techniques,” in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025, pp. 1–6. DOI: [10.1109/ECCE64574.2025.11013062](https://doi.org/10.1109/ECCE64574.2025.11013062)
2. **A. B. Rahman**, M. Touhid Islam, M. R. Islam, M. Sohrawordi, and M. N. Sultan, “Enhanced Brain Tumor Classification from MRI Images Using Deep Learning Model,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 2023, pp. 1–6. DOI: [10.1109/ICCIT60459.2023.10441064](https://doi.org/10.1109/ICCIT60459.2023.10441064)

Preprints

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Deep Learning Architectures for Medical Image Denoising: A Comparative Study of CNN-DAE, CADTra, and DCMIEDNet*, 2025. arXiv: [2508.17223](https://arxiv.org/abs/2508.17223) [eess.IV]. URL: <https://arxiv.org/abs/2508.17223>

2. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Systematic Evaluation of Wavelet-Based Denoising for MRI Brain Images: Optimal Configurations and Performance Benchmarks*, 2025. arXiv: [2508.15011](https://arxiv.org/abs/2508.15011) [eess.IV].  URL: <https://arxiv.org/abs/2508.15011>

Teaching

- | | |
|-------------|--|
| Summer 2026 | <ul style="list-style-type: none">◇ Course Teacher, WUB. Compiler Design.◇ Course Teacher, WUB. Object Oriented Programming.◇ Course Teacher, WUB. Introduction to Programming. |
| Spring 2026 | <ul style="list-style-type: none">◇ Course Teacher, WUB, Object Oriented Programming.◇ Course Teacher, WUB. Introduction to Programming. |
| Fall 2025 | <ul style="list-style-type: none">◇ Course Teacher, WUB. Computer Fundamentals.◇ Course Teacher, WUB. Object Oriented Programming.◇ Course Teacher, WUB. Computer and Cyber Security. |
| Summer 2023 | <ul style="list-style-type: none">◇ Instructor, SASTC. Theory of Computation and Concrete Mathematics.◇ Instructor, SASTC. Computer Graphics and Image Processing.◇ Instructor, SASTC. Machine Learning and Pattern Recognition. |

Research and Projects

- | | |
|-------|--|
| 02/26 | ◇ Multilayered Security  : Implemented and simulated a research project on Multi-Layered Security System: Integrating Quantum Key Distribution with Classical Cryptography to Enhance Steganographic Security. |
| 11/24 | ◇ Guava Fruit Disease Dataset @ IoThink Lab  : Collaborated on dataset collection for interdisciplinary research. |
| 08/24 | <ul style="list-style-type: none">◇ Quantum Variational Classifier @ Womanium Program : Implemented a quantum classifier for Penguin Species Classification.◇ Quantumvolutional Neural Networks @ Womanium Program : Developed a hybrid quantum convolutional model for MNIST Digit Classification.◇ Quantum Regression Model @ Womanium Program : Implemented a Quantum Machine Learning Model to learn and predict the sine function on the interval $[0, 2\pi]$.◇ Quantum Machine Learning for Anomaly Detection @ Womanium Program : Developed a hybrid model for anomaly detection, leading to project finalist recognition. |

Professional Development

- | | |
|---------------|---|
| 09/24 – 12/24 | ◇ QClass24/25 Fall Semester  : Achieved 96% in 3 ECTS graduate-level program on Quantum Algorithms and QKD. |
| 06/24 – 08/24 | ◇ Womanium Quantum + AI 2024  : Coursework on Quantum Computing and AI. |
| 08/24 | <ul style="list-style-type: none">◇ QGSS 2024 – Quantum Excellence  : Intensive quantum computing boot camp by IBM Qiskit.◇ Classiq Diploma : Advanced Quantum Algorithm design.◇ Pennylane Diploma : Quantum Machine Learning Challenge completion. |
| 07/24 | <ul style="list-style-type: none">◇ QNickel Diploma : Quantum Algorithms workshop.◇ QBronze Diploma : Introductory Quantum Computing and Programming. |
| 09/20 | ◇ Python for Everybody Specialization  : Coursera specialization on Python fundamentals and data structures. |

Awards and Achievements

- 06/25 ◇ **Unitary Hack 2025**    : For open source contributions to the quantum software ecosystem.
- 04/25 ◇ **YQuantum 2025**   : Solved BlueQubit's Peaked Circuits Challenge.
- 02/25 ◇ **MIT iQuHACK 2025**  : Solved IonQ's Max Cut problem.
- 08/24 ◇ **Womanium Quantum + AI 2024**  : Program finalist and QSL fellowship nominee.
- 06/24 ◇ **IBM Quantum Challenge 2024**   : Ranked 11th globally.
- 01/16 ◇ **6th BAS Science Olympiad**  : Division 3rd and National level nominee.
- 09/15 ◇ **5th DSA Physics Olympiad**  : 7th Place.

Volunteering Work

- 04/25 – Present ◇ **Coordinator & Mentor, QBangladesh (QWorld)**  : Mentoring students and promoting quantum computing literacy in Bangladesh.
Hosted Events: QBronze , QSilver .

Hobbies

- ◇ Chess 
- ◇ Gardening

References

Dr. Md. Abdulla Al Mamun

Professor

Department of Computer Science and Engineering
Hajee Mohammad Danesh Science and Technology University (HSTU)

Phone: +8801886890345

Email: mamun@hstu.ac.bd

Dr. Masud Ibn Afjal

Professor

Department of Computer Science and Engineering
Hajee Mohammad Danesh Science and Technology University (HSTU)

Phone: +8801737049633

Email: masud@hstu.ac.bd